

DATA 581

Modelling and Simulation II



Continuous-Time Markov Chains

Outline

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 1. M/M/1 Queue (as a Continuous-Time Markov Chain)
 2. Queueing Models as Discrete-Time Markov Chains
 3. Little's Formula
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```
set.seed(222696) # use this to reproduce output
```

Some Properties of Exponential Random Variables

Memoryless Property:

If X is an exponential random variable, then for any $r, s \geq 0$,

$$P(X > r + s | X > r) = P(X > s).$$

e.g. Consider an exponential with $\lambda = 2$, and take $r = 1.5$ and $s = 0.75$:

```
N <- 1000000 # number of simulated values
r <- 1.5; s <- 0.75
X <- rexp(N, rate = 2)
mean(X[X > r] > (r + s)) # P(X > r+s | X > r)

## [1] 0.222446

mean(X > s) # P(X > s)

## [1] 0.222617
```

Some Properties of Exponential Random Variables

Minimum Property I:

If X and Y are independent exponential random variables with means $1/\lambda$ and $1/\mu$, respectively. Then

$$P(\min(X, Y) \geq t) = e^{-(\mu+\lambda)t}$$

That is, $\min(X, Y)$ is an exponential random variable with mean $\frac{1}{\mu+\lambda}$.

e.g. with $\lambda = 2, \mu = 3$:

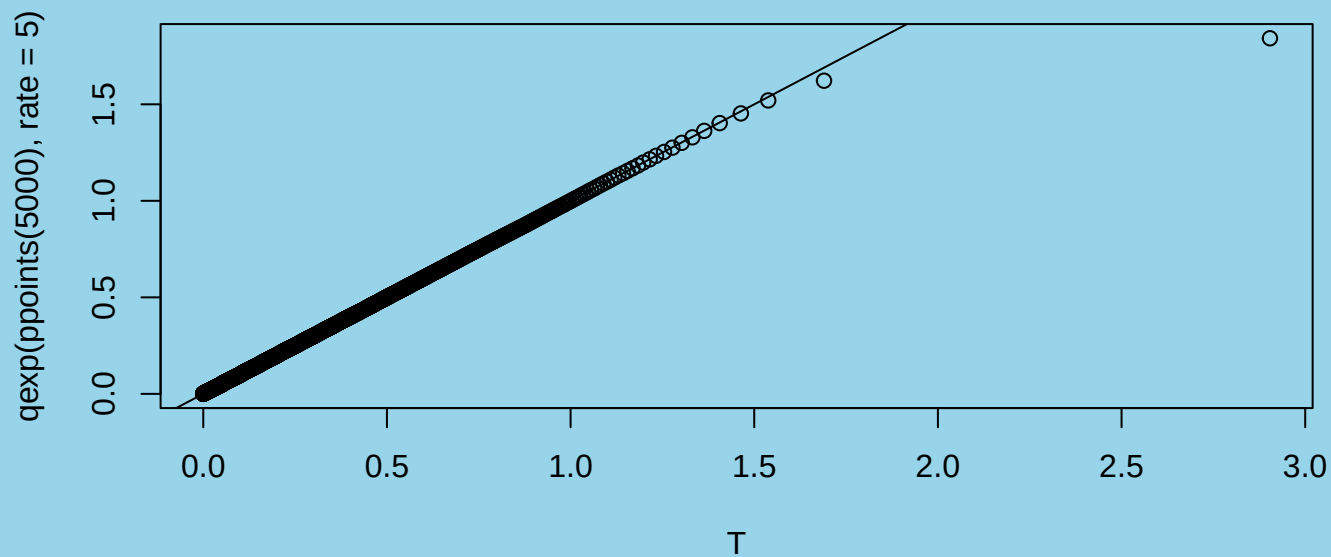
```
Y <- rexp(N, rate = 3);    T <- pmin(X, Y)  # pairwise min
mean(T) # compare with 1/5

## [1] 0.1998387
```

Some Properties of Exponential Random Variables

An exponential QQ-plot of T:

```
par(mar=c(4, 4, 1, 1))
qqplot(T, qexp(ppoints(5000), rate = 5))
abline(0, 1)
```



Some Properties of Exponential Random Variables

Minimum Property II:

If X and Y are independent exponential random variables with means $1/\lambda$ and $1/\mu$, respectively, then

$$P(X < Y) = \frac{\lambda}{\lambda + \mu}$$

e.g.

```
mean(X < Y) # compare with 2/5  
## [1] 0.400035
```

The General Framework

We define $X(t)$ as a continuous time Markov chain, if it is a random function of t which is piecewise constant with jumps at random times ... which will be defined presently.

There exist constants q_{jj} and q_{ij} such that for very small Δ , the following hold:

$$P(X(t + \Delta) = j | X(t) = j) = 1 - \Delta q_{jj} + o(\Delta)$$

$$P(X(t + \Delta) = j | X(t) = i) = \Delta q_{ij} + o(\Delta)$$

The term $o(\Delta)$ is very small – if you divide by Δ and let Δ tend to 0, you still get 0.

The General Framework

The first equation essentially says that there is a very high probability of no jump within a short time interval. The second equation says that there is a small probability of a jump from state i to state j in a short time interval.

Let $p_j(t) = P(X(t) = j)$ for $j \in \{1, 2, \dots, K\} = S$, the state space. Then

$$p_j(t+\Delta) = P(X(t+\Delta) = j) = \Delta \sum_{i \neq j} q_{ij} p_i(t) - \Delta q_{jj} p_j(t) + p_j(t) + o(\Delta).$$

Subtracting $p_j(t)$ from both sides, dividing by Δ and taking the limit as $\Delta \rightarrow 0$ gives

$$p'_j(t) = \sum_{i \neq j} q_{ij} p_i(t) - q_{jj} p_j(t).$$

The General Framework

If we define the *infinitesimal generator* Q as the $K \times K$ matrix where $Q_{jj} = -q_{jj}$ and $Q_{ij} = q_{ij}$ for $i \neq j$, we can write

$$P'(t) = P(t)Q$$

where $P(t)$ is a K -vector of the probabilities $p_j(t)$, $j = 1, 2, \dots, K$.

In fact, $P(t)$ could be taken to be a $K \times K$ matrix itself, if we want to take into account the initial value of the Markov chain $X(0)$.

The solution to the matrix differential equation is

$$P(t) = P(0)\mathbf{e}^{Qt}.$$

The General Framework

If the Markov chain has a steady state distribution, the probability distribution should become constant eventually. This means that the derivative of $P(t)$ must become 0.

Therefore, the steady state distribution π , if it exists, satisfies

$$0 = \pi Q$$

Therefore, we can find the steady state distribution from the above equation, with the additional condition that the probabilities sum to 1.

The General Framework

What are the q_{ij} and q_{jj} values?

The q_{ij} values are determined as exponential rates of transition from state i to state j .

The q_{jj} values represent the time that the Markov chain stays in state j before making any other transition.

From the fact that $P'(t) = P(t)Q$, we have that $P(t)Q\mathbf{1} = 0$, since the derivative of the sum of the probabilities must be 0. This implies that $Q\mathbf{1} = 0$. That is the row sums of Q must be 0.

Therefore, $q_{jj} = -\sum_{j \neq i} q_{ji}$

A Simple Continuous Time Markov Chain

Consider a light bulb which lasts for an exponential λ_0 amount of time before burning out. It is replaced in an exponential λ_1 amount of time, before burning out again, and so on.

Let

$$X(t) = \begin{cases} 1, & \text{if the bulb works at } t \\ 0, & \text{if the bulb is being replaced at } t \end{cases}$$

and let $P_{0i}(t) = P_0(X(t) = i)$, for $t \geq 0, i = 0, 1$.

It follows that $P_{00}(0) = 1$, and $P_{01}(0) = 0$.

A Simple Continuous Time Markov Chain

Goal: to calculate $P_{00}(t)$ for any $t > 0$.

Using the memoryless property of the exponential distribution, we can show that

$$P'_{00}(t) = -\lambda_0 P_{00}(t) + \lambda_1 P_{01}(t).$$

Similarly,

$$P'_{01}(t) = \lambda_0 P_{00}(t) - \lambda_1 P_{01}(t).$$

Therefore, we can write

$$P'(t) = PQ$$

where

$$Q = \begin{bmatrix} -\lambda_0 & \lambda_0 \\ \lambda_1 & -\lambda_1 \end{bmatrix} \quad \text{and} \quad P(t) = [P_{00}(t) \ P_{01}(t)].$$

A Simple Continuous Time Markov Chain

The solution of this matrix differential equation is

$$P(t) = P(0)e^{Qt}.$$

If $Q = XDX^{-1}$ for some diagonal matrix D , then

$$P(t) = P(0)Xe^{Dt}X^{-1}.$$

In this case,

$$X = \begin{bmatrix} 1 & \lambda_0/\lambda_1 \\ 1 & 1 \end{bmatrix}$$

and

$$D = \begin{bmatrix} 0 & 0 \\ 0 & -\lambda_0 - \lambda_1 \end{bmatrix}$$

A Simple Continuous Time Markov Chain

We conclude that

$$P_{00}(t) = \frac{\lambda_1 + \lambda_0 e^{-(\lambda_0 + \lambda_1)t}}{\lambda_1 + \lambda_0}$$

If $t \rightarrow \infty$, we have

$$q_0 = \frac{\lambda_1}{\lambda_0 + \lambda_1}$$

The limiting steady-state probability vector q can more simply be obtained by solving

$$qQ = 0 \quad q1 = 1$$

A Compartmental Model*

Suppose an individual can be in one of 4 states: Healthy, Disease₁, Disease₂ and Disease₃. None of the diseases is immediately fatal and it is possible to recover one's health from any of the three diseases.

We suppose that the individual can transit from the healthy state to Disease₁ at a rate of 0.1 and to Disease₂ with a rate of 0.2. They do not transit directly to Disease₃.

They can recover from Disease₁ at a rate of 0.9 or progress to Disease₃ at a rate of 0.1. From Disease₂ they can recover at a rate of .4 and progress to Disease₃ at a rate of 0.6. From Disease₃, they can recover at a rate of 0.1.

Write down the Q matrix for this model and find the stationary vector.

*See also Kucharski et al (2020), Early dynamics of transmission and control of COVID-19: a mathematical modelling study. *Lancet*. In that study, the states are susceptible, diseased, recovered and dead. The last two states are called absorbing states and the rates out of the those states are 0.

A Compartmental Model

$$Q = \begin{bmatrix} -.3 & .1 & .2 & 0 \\ .9 & -1 & 0 & .1 \\ .4 & 0 & -1 & .6 \\ .1 & 0 & 0 & -.1 \end{bmatrix}.$$

```
Q <- matrix(c(-.3, .9, .4, .1, .1, -1, 0, 0, .2, 0,
              -1, 0, 0, .1, .6, -.1), nrow=4)
QT <- rbind(t(Q), rep(1, 4)) # include a row of 1's
RHS <- c(rep(0, 4), 1) # equate Q-transpose times pi to 0
qr.solve(QT, RHS) # solve for stationary distribution

## [1] 0.38461538 0.03846154 0.07692308 0.50000000
```

A Compartmental Model

Find the vector of probabilities of being in any of the 4 states at a given time, assuming that the individual starts in the healthy state.

```
Q <- matrix(c(-.3, .9, .4, .1, .1, -1, 0, 0, .2, 0, -1, 0,
              0, .1, .6, -.1), nrow=4)
evals <- eigen(Q)$values
evals

## [1] -1.179583e+00 -1.000000e+00 -2.204168e-01 -5.551115e-17

X <- eigen(Q)$vector
X

##           [,1]           [,2]           [,3] [,4]
## [1,]  0.18477758 -1.045612e-16 -0.5972189 -0.5
## [2,] -0.91650166  8.944272e-01 -0.6258486 -0.5
## [3,] -0.35438540 -4.472136e-01  0.0752815 -0.5
## [4,] -0.01711564 -2.537681e-16  0.4959596 -0.5
```

A Compartmental Model

```
P <- function(t) {
  Pmat <- matrix(0, nrow=length(t), ncol=4)
  for (i in 1:length(t)) {
    Pmat[i, ] <- exp(evals*t[i])%*%diag(X[1,])%*%solve(X)
  }
  Pmat
}
```

The result is the probabilities which have been calculated as $P(0)\mathbf{e}^{Qt} = P(0)X\mathbf{e}^{Dt}X^{-1}$.
 The first one (for the healthy state) is $P_1(t) = 0.171\mathbf{e}^{-1.18t} + 0.444\mathbf{e}^{-0.22t} + 0.385$.

A Compartmental Model

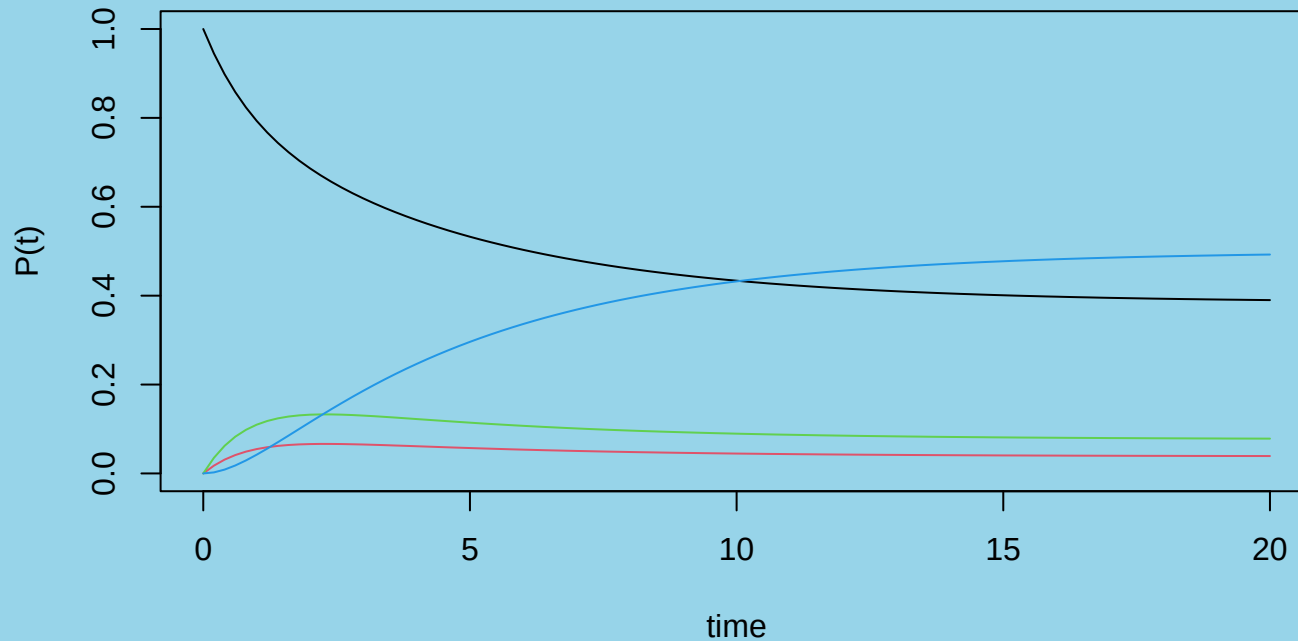
Evaluate the probabilities at times 1 and 2, and plot the probabilities as functions of time over the interval from 0 through 20.

```
P(c(1, 2))
```

```
##           [, 1]      [, 2]      [, 3]      [, 4]
## [1, ] 0.793 0.0548 0.110 0.0421
## [2, ] 0.687 0.0661 0.132 0.1152
```

A Compartmental Model

```
par(mar=c(4, 4, 1, 1))
curve(P(x)[,1], 0, 20, xlab="time", ylim=c(0, 1), ylab="P(t)")
for (i in 2:4) {
  curve(P(x)[,i], 0, 20, add = TRUE, col=i)
}
```



Notice that these curves all heading towards the stationary distribution calculated on slide 17.

The Linear Birth Process

In this process, $X(t)$ counts up the number of organisms alive at time t . $X(0)$ is the initial population size and growth of the population increases by 1 each time one of the organisms divides into two offspring. Each living organism does this after being alive for an exponential λ amount of time.

For this process, the Q matrix is infinite with nonzero entries at $(j - 1, j)$ and (j, j) only: $q_{j-1,j} = \lambda(j - 1)$ and $q_{j,j} = -\lambda j$.

It is then possible to show that

$$p'_1(t) = -\lambda p_1(t) \quad \text{and}$$

$$p'_j(t) = \lambda(j - 1)p_{j-1}(t) - \lambda j p_j(t).$$

The Linear Birth Process

It is easy to solve the first equation to see that

$$p_1(t) = p_1(0)\mathbf{e}^{-\lambda t}.$$

If the initial population size is 1, then $p_1(0) = 1$. The rest of the equations can be solved sequentially, inductively, to obtain

$$p_k(t) = \mathbf{e}^{-\lambda t}(1 - \mathbf{e}^{-\lambda t})^{k-1}, \quad \text{for } k = 2, 3, \dots$$

From this, or from the differential equations directly, we can obtain

$$E[X(t)] = X(0)\mathbf{e}^{\lambda t}$$

This is the classical exponential growth model.

Another Look at the Poisson Process

Other models for the Q matrix are possible. For example, suppose $q_{j,j} = -\lambda$ and $q_{j-1,j} = \lambda$, i.e. constant, instead of linearly depending on j , and that $p_0(0) = 1$.

It is then possible to show that

$$p'_0(t) = -\lambda p_0(t) \quad \text{and}$$

$$p'_j(t) = \lambda p_{j-1}(t) - \lambda p_j(t).$$

The solution to the first equation, with the initial condition is $p_0(t) = e^{-\lambda t}$ which then can be used to begin to sequentially solve the other equations yielding

$$p_j(t) = \frac{(\lambda t)^j e^{-\lambda t}}{j!}, j = 0, 1, 2, \dots$$

In other words, the probability of j points in the interval $[0, t]$ is the Poisson probability with rate λt .

The Linear Birth and Death Process

Now, suppose that organisms can die as well, and that they live for an exponential μ amount of time before dying. They still divide after an exponential λ amount of time as well. (The memoryless property of exponential random variables helps to resolve the apparent contradiction.)

The equations for the probabilities are:

$$p'_j(t) = \lambda(j-1)p_{j-1}(t) + \mu(j+1)p_{j+1}(t) - j(\lambda + \mu)p_j(t).$$

Solving these equations is possible, using Laplace transforms $\rightsquigarrow$ beyond the scope of this course.

We could also construct equations for nonlinear birth and death processes, by replacing the $j+1$ and $j-1$ terms with functions of $j+1$ and $j-1$, such as square roots or reciprocals and so on.

The Linear Birth and Death Process

Finding the mean population size is not difficult.

Let $x(t) = E[X(t)]$ and multiply the above equations by j and sum over all j to obtain

$$x'(t) = (\lambda - \mu)x(t)$$

which has the solution $x(t) = e^{(\lambda - \mu)t}$.

If $\lambda > \mu$, the population grows exponentially. If $\lambda < \mu$, the population decays exponentially, and dies out. If $\lambda = \mu$, the population also dies out, but the mean population size is $x(0)$ for all t .

Queueing Systems

Queues (line-ups) were first studied when telephone networks were being developed in the early 20th century.

Telephones were connected to networks of exchanges .

Queues of customers formed at exchanges, waiting for a circuit.

Problems to solve: what configuration of switches and exchanges were needed to handle the traffic? Key performance indicators include:

- expected queue-length**
- expected waiting time in queue**
- expected time of a customer in the system**
- probability that the waiting time exceeds a given amount**
- probability that the queue length exceeds a given requirement**

The M/M/1 Queue

This queue is a simple birth and death process where the birth and death rates are always the same: λ and μ . Customers arrive at the queue according to a Poisson process with rate λ and they are served according to an exponential distribution with rate μ .

The equations for the state probabilities are

$$p'_j(t) = \lambda p_{j-1}(t) + \mu p_{j+1}(t) - (\lambda + \mu) p_j(t)$$

and

$$p'_1(t) = -\lambda p_0(t) + \mu p_1(t).$$

Again, transform methods can be used to solve for the state probabilities.

The M/M/1 Queue

We can find the steady state probability vector, when $\lambda < \mu$:

$$\lambda p_{j-1}(t) + \mu p_{j+1}(t) - (\lambda + \mu)p_j(t) \rightarrow 0$$

as $t \rightarrow \infty$, so the steady state probabilities π_j satisfy

$$\lambda \pi_{j-1} + \mu \pi_{j+1} - (\lambda + \mu)\pi_j = 0$$

and

$$-\lambda \pi_0 + \mu \pi_1 = 0.$$

The M/M/1 Queue

The last equation says that $\pi_1 = \lambda\pi_0/\mu$. We can solve the other equations for π_j in terms of π_0 . Then we use the fact that the probabilities must sum to 1, to see that

$$\pi_0 = 1 - \frac{\lambda}{\mu}$$

and

$$\pi_k = \left(\frac{\lambda}{\mu}\right)^k \left(1 - \frac{\lambda}{\mu}\right)$$

for $k = 0, 1, 2, \dots$

$M/G/1$ Model

Customers arrive for service according to a Poisson process with rate λ .

Service time is randomly distributed according to a probability distribution $G(t) = P(S \leq t)$, with probability density function $g(t) = G'(t)$.

There is only one server (generalization: $M/G/m$ has m servers) so when there is more than one customer, a queue forms.

This kind of queueing system can be analyzed as a continuous-time Markov chain, but a lot of the necessary information can be obtained by treating it as a discrete time Markov chain.

Let X_n denote the number of customers in the system (in queue plus in service) immediately after the n th customer has completed service. $\{X_n\}$ is now a discrete time Markov chain.

Transition Matrix for $M/G/1$ Model

From the time the n th customer enters service until service completion is a random variable T_n .

The number of customers joining the queue during this period is $N(T_n)$ where N is a Poisson process with parameter λ .

Given T_n , the number of customers arriving is Poisson distributed with parameter λT_n .

$$P(N(T_n) = j | T_n = t) = \frac{e^{-\lambda t} (\lambda t)^j}{j!}, \quad j = 0, 1, 2, \dots$$

Transition Matrix for $M/G/1$ Model

By the Law of Total Probability, we have

$$\begin{aligned} p_j &:= P(N(T_n) = j) = \int_0^\infty P(N(T_n) = j | T_n = t) g(t) dt \\ &= \int_0^\infty \frac{e^{-\lambda t} (\lambda t)^j g(t)}{j!} dt. \end{aligned}$$

Then, if $X_n \geq 1$,

$$X_{n+1} = X_n + j - 1, \quad \mathbf{w.p.} \ p_j, j = 1, 2, \dots$$

Transition Matrix for $M/G/1$ Model

Note that if $X_n = 0$, we need to wait for a customer to arrive; that customer immediately enters service.

Therefore, the time until the next service completion when $X_n = 0$ is $T_n + E_n$, where E_n is exponentially distributed with parameter λ .

That means that the number of arrivals until the next service completion is also $N(T_n)$, and the transitions from state 0 to state j are the same as from state 1 to state j .

Transition Matrix for $M/G/1$ Model

$$P = \begin{bmatrix} p_1 & p_2 & p_3 & \cdots & \cdots \\ p_1 & p_2 & p_3 & \cdots & \cdots \\ 0 & p_1 & p_2 & \cdots & \cdots \\ 0 & 0 & p_1 & \cdots & \cdots \\ . & . & . & \cdots & \cdots \end{bmatrix}$$

Simulating a Special Case

We could calculate the integrals to obtain the p 's, or we can model the p 's themselves directly:

e.g.

$$p_j = (1 - p)^{j-1}p$$

for some μ and some p . (p_j follows a geometric probability model and can be simulated using `rgeom()`)

Simulating a Special Case

```
p <- 0.7; current.state <- 0; Ntransitions <- 10000
X <- numeric(Ntransitions)
for (i in 1:Ntransitions) {
  if (current.state == 0) {
    current.state <- rgeom(1, prob = p)
  } else {
    transition <- rgeom(1, prob = p)
    current.state <- current.state + transition - 1
  }
  X[i] <- current.state
}
table(X)
```

```
## X
##      0      1      2      3      4      5      6      7      8      9     10
## 5793 2419  974  444  200   92   46   19    4    5    4
```

Little's Formula

The long-run average number L of customers in a stationary system is equal to the long-run average arrival rate λ multiplied by the average time W that a customer spends in the system.

$$L = \lambda W.$$

We can use this formula, together with the simulation results, to estimate the mean time a customer is in the system, given knowledge of the arrival rate, e.g. $\lambda = 20/\text{minute}$:

```
lambda <- 20  
W <- mean(X) / lambda * 60 # seconds  
W  
  
## [1] 2.25
```

Simulating another Special Case

```
p <- 0.51; current.state <- 0; Ntransitions <- 10000
X <- numeric(Ntransitions)
for (i in 1:Ntransitions) {
  if (current.state == 0) {
    current.state <- rgeom(1, prob = p)
  } else {
    transition <- rgeom(1, prob = p)
    current.state <- current.state + transition - 1
  }
  X[i] <- current.state
}
W <- mean(X) / lambda * 60 # seconds
W

## [1] 61
```

Simulating yet another Special Case

```
p <- 0.5
current.state <- 0
Ntransitions <- 10000
X <- numeric(Ntransitions)
for (i in 1:Ntransitions) {
  if (current.state == 0) {
    current.state <- rgeom(1, prob = p)
  } else {
    transition <- rgeom(1, prob = p)
    current.state <- current.state + transition - 1
  }
  X[i] <- current.state
}
summary(X)
```

##	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
##	0	56	95	93	124	192

... instability ... steady state distribution does not exist ... service is too slow

and Little's formula does not apply, since the process is not stationary

Another Special Case: $M/M/1$ Model

Customers arrive for service according to a Poisson process with rate λ .

Service time is randomly distributed according to an exponential probability distribution with rate μ .

There is only one server (generalization: $M/M/m$ has m servers) so when there is more than one customer, a queue (line-up) forms.

Let X_n denote the number of customers in the system (in queue plus in service) immediately after the next event – either the next arrival of a customer or the next customer has completed service.

$M/M/1$ Queue

The only possible transitions are to increase by 1 or to decrease by 1.

The probability of increasing the queue length by 1: $\frac{\lambda}{\mu + \lambda} = p$

(i.e. customer arrives before service completion)

The probability of decreasing the queue length by 1: $\frac{\mu}{\mu + \lambda} = 1 - p$

Transition Matrix for $M/M/1$ Model:

$$P = \begin{bmatrix} 0 & 1 & 0 & 0 & \dots \\ 1 - p & 0 & p & 0 & \dots \\ 0 & 1 - p & 0 & p & \dots \\ 0 & 0 & 1 - p & 0 & \dots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

Simulating a Special Case

```
p <- 0.3; current.state <- 0; Ntransitions <- 10000
X <- numeric(Ntransitions)
for (i in 1:Ntransitions) {
  if (current.state == 0) {
    current.state <- 1
  } else {
    transition <- sample(c(-1,1), prob=c(1-p, p), size = 1)
    current.state <- current.state + transition
  }
  X[i] <- current.state
}
table(X)
```

##	X	0	1	2	3	4	5	6	7	8	9
##		2794	4044	1813	779	325	150	61	25	7	2

Simulating a Special Case

```
p <- 0.51
current.state <- 0
Ntransitions <- 10000
X <- numeric(Ntransitions)
for (i in 1:Ntransitions) {
  if (current.state == 0) {
    current.state <- 1
  } else {
    transition <- sample(c(-1,1), prob=c(1-p, p), size = 1)
    current.state <- current.state + transition
  }
  X[i] <- current.state
}
summary(X)
```

##	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
##	0.0	24.0	42.0	49.1	69.0	123.0

... unstable again ... traffic intensity is $.51/.49 > 1$...

Steady State Distribution

$$\pi_0 = \frac{1 - 2p}{2(1 - p)}$$

$$\pi_j = \pi_0 \frac{p^{j-1}}{(1 - p)^j}, \text{ for } j \geq 1$$

Note that this only makes sense when $p < .5$. When $p \geq .5$, the queue is unstable.

Summary

Continuous-time Markov chains are the stochastic starting point to differential equation which comprise an enormous class of models for physical and biological phenomena.

Under similar conditions as for discrete-time Markov chains, steady state distributions can be obtained by solving a linear system of equations.

Compartmental models, multi-state and queueing models are important special cases.

Simulating from these models is similar to simulating from discrete time Markov chains, if one exploits the special properties of the exponential distribution.

Relaxing the exponential rate assumption leads to processes called Semi-Markov processes which are more realistic but much more difficult to simulate.